CLASS RANKING SYSTEM

```java

public class ClassRankingSystem {

    // Student class
    static class Student {
        String name;
        int[] quizzes;

        Student(String name, int[] quizzes) {
            this.name = name;
            this.quizzes = quizzes;
        }

        // Calculate the average quiz score
        double getAverage() {
            int sum = 0;

            for (int i = 0; i < quizzes.length; i++) {
                sum += quizzes[i];
            }

            return (double) sum / quizzes.length;
        }
    }

    // Custom bubble sort: highest average comes first
    static void sortStudents(Student[] students) {
        for (int i = 0; i < students.length - 1; i++) {
            for (int j = 0; j < students.length - 1 - i; j++) {

                if (students[j].getAverage()
                        < students[j + 1].getAverage()) {

                    Student temp = students[j];
                    students[j] = students[j + 1];
                    students[j + 1] = temp;
                }
            }
        }
    }

    // Linear search that returns the student's rank
    static int linearSearchRank(Student[] students,
                                String searchName) {

        for (int i = 0; i < students.length; i++) {

            if (students[i].name.equalsIgnoreCase(searchName)) {

                // Count students with a higher average.
                int rank = 1;

                for (int j = 0; j < students.length; j++) {
                    if (students[j].getAverage()
                            > students[i].getAverage()) {
                        rank++;
                    }
                }

                return rank;
            }
        }

        return -1; // Student not found
    }

    public static void main(String[] args) {

        Student[] students = {
            new Student("Zea", new int[]{90, 95, 88, 92}),
            new Student("Maine", new int[]{85, 80, 90, 85}),
            new Student("Marga", new int[]{95, 92, 96, 93}),
            new Student("Rodge", new int[]{85, 80, 90, 85}),
            new Student("Jov", new int[]{78, 82, 80, 84})
        };

        // Sort students by average, highest to lowest.
        sortStudents(students);

        System.out.
println("=========================================");
        System.out.println("         CLASS RANKING SYSTEM");
        System.out.
println("=========================================");
        System.out.printf("%-8s %-12s %-10s%n",
                "Rank", "Name", "Average");
        System.out.println("-----------------------------------");

        for (int i = 0; i < students.length; i++) {

            // Equal averages receive the same rank.
            int rank = 1;

            for (int j = 0; j < students.length; j++) {
                if (students[j].getAverage()
                        > students[i].getAverage()) {
                    rank++;
                }
            }

            System.out.printf("%-8d %-12s %-10.2f%n",
                    rank,
```

```
81        System.out.println("          CLASS RANKING SYSTEM");
82        System.out.
    println("========================================");
83        System.out.printf("%-8s %-12s %-10s%n",
84              "Rank", "Name", "Average");
85        System.out.println("----------------------------------------");
86
87        for (int i = 0; i < students.length; i++) {
88
89            // Equal averages receive the same rank.
90            int rank = 1;
91
92            for (int j = 0; j < students.length; j++) {
93                if (students[j].getAverage()
94                        > students[i].getAverage()) {
95                    rank++;
96                }
97            }
98
99            System.out.printf("%-8d %-12s %-10.2f%n",
100                 rank,
101                 students[i].name,
102                 students[i].getAverage());
103       }
104
105       System.out.
    println("========================================");
106
107       // Search for a student and display their rank.
108       String searchName = "";
109       int result = linearSearchRank(students, searchName);
110
111       System.out.println("\n=== STUDENT RANK SEARCH ===");
112       System.out.println("Searching for: " + searchName);
113
114       if (result != -1) {
115           System.out.println(searchName + "'s rank is: "
116                   + result);
117       } else {
118           System.out.println("Student not found.");
119       }
120   }
121 }
```

```
========================================
          CLASS RANKING SYSTEM
========================================
Rank     Name         Average
----------------------------------------
1        Marga        94.00
2        Zea          91.25
3        Maine        85.00
3        Rodge        85.00
5        Jov          81.00
========================================

=== STUDENT RANK SEARCH ===
Searching for:
Student not found.

[Program finished]
```